

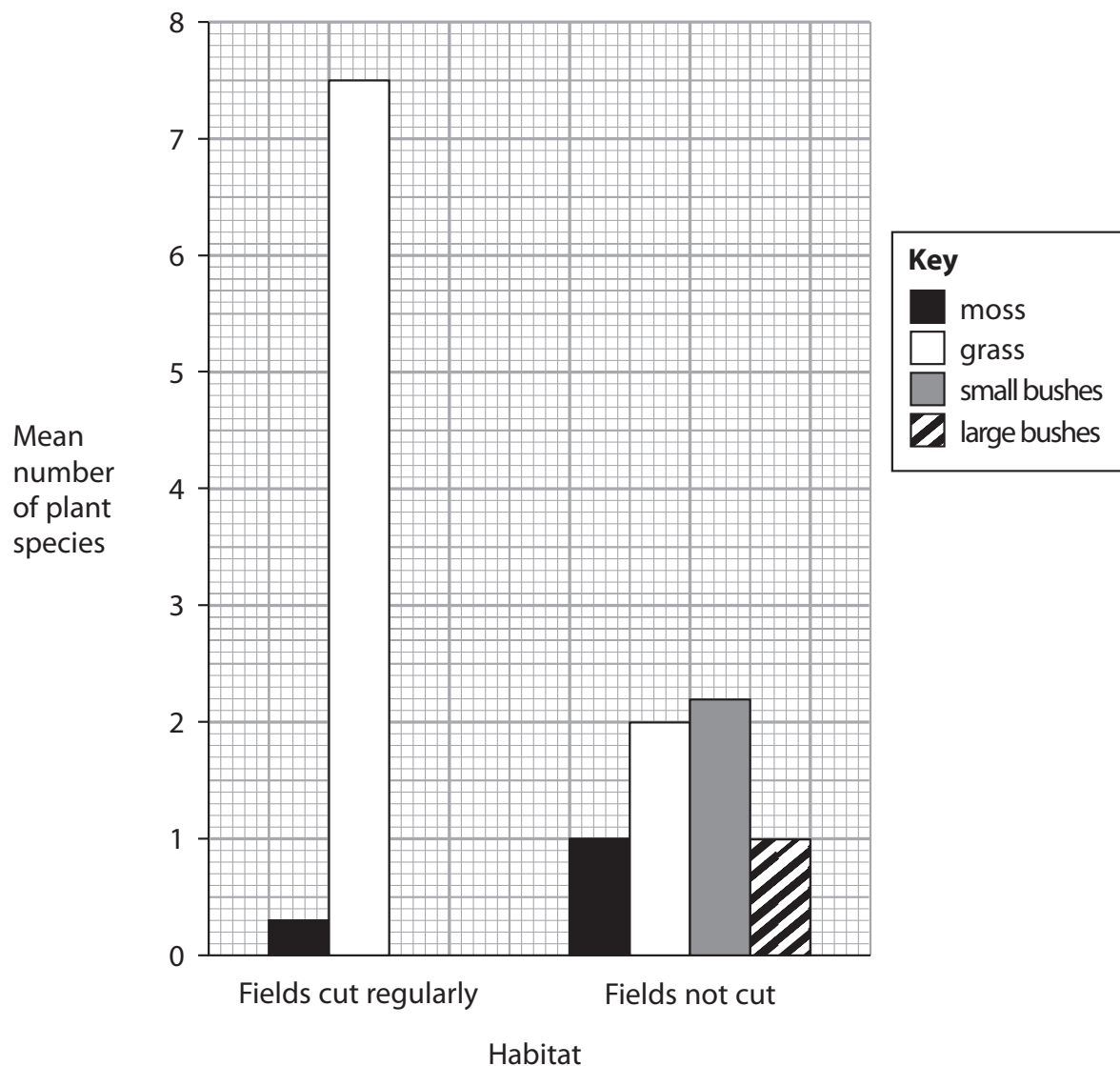
1 Mosses are species of plants that have genomes that enable them to survive in their habitats.

(a) A group of students investigated the mean number of plant species in four habitats.

The habitats that they investigated were:

- fields that were cut regularly
- fields that were not cut
- the middle of woods
- the edge of woods.

The graph shows some of the results from this investigation.



- (i) Which row of the table shows the type of factor that affected the plant species in two of these habitats?

(1)

	Cutting the field regularly	Competition with trees for water
☐ A	abiotic factor	abiotic factor
☐ B	abiotic factor	biotic factor
☐ C	biotic factor	abiotic factor
☐ D	biotic factor	biotic factor

- (ii) What is the percentage increase in the number of species of moss between the fields cut regularly and the fields not cut?

(1)

- ☐ A 23.3
- ☐ B 70.0
- ☐ C 233.3
- ☐ D 333.3

- (iii) Suggest reasons for the differences in the mean number of plant species in fields cut regularly and in fields not cut.

(2)

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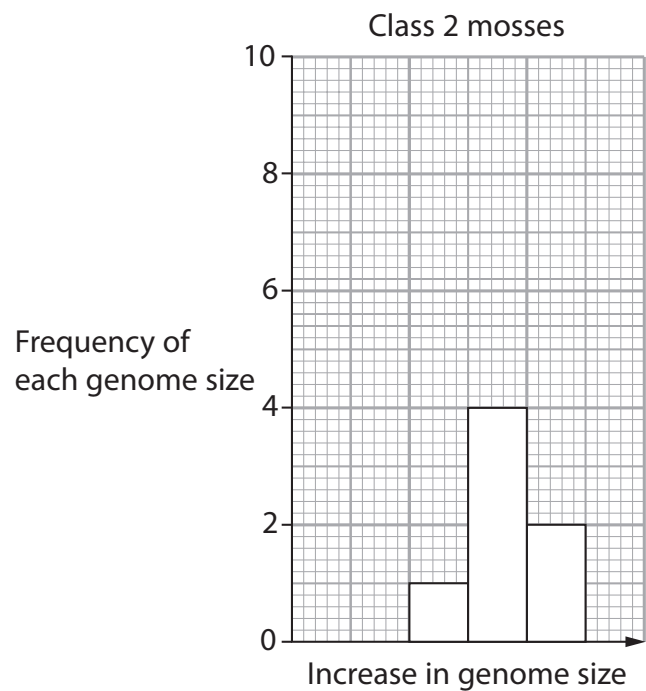
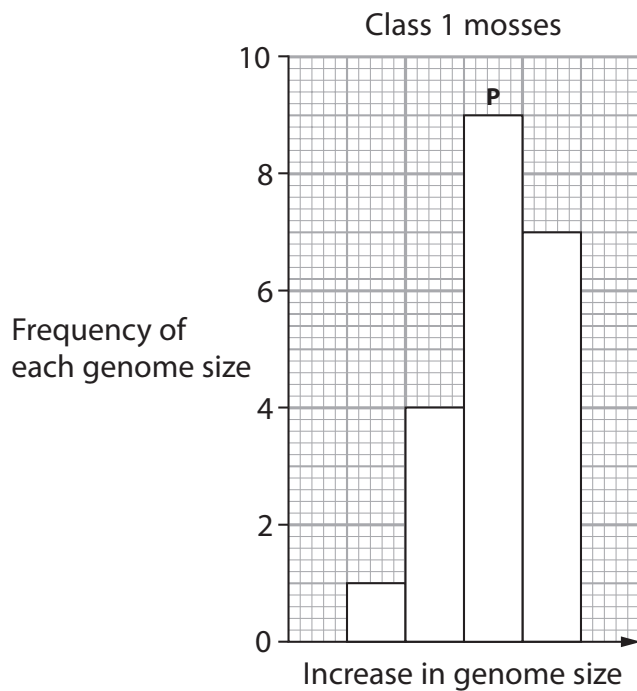
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- (b) In another investigation, the genome size of 30 species of moss was determined. Each species of moss belonged to one of two classes of moss.

The graphs show the frequency of each genome size in the class 1 mosses and some of the class 2 mosses.



- (i) Which does **P** represent?

(1)

- ☐ **A** the maximum genome size on a bar chart
- ☐ **B** the maximum genome size on a histogram
- ☐ **C** mode genome size on a bar chart
- ☐ **D** mode genome size on a histogram

- (ii) One genome size is missing from the graph for class 2 mosses.

Calculate the frequency of this genome size.

(1)

Answer.....

- (iii) The mean value for the genome size of class 1 mosses was 0.449 a.u. and the mean value for class 2 mosses was 0.920 a.u.

Calculate the ratio of the genome size of class 1 mosses to the genome size of class 2 mosses.

(1)

Answer.....

- (iv) It was suggested that the chromosomes in class 2 mosses were found in pairs.

Give the evidence that supports this suggestion.

(1)

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(Total for Question 1 = 8 marks)